Experiment- 5

OBJECT: Study the construction and connection of single-phase transformer and auto-transformer. Measure input and output voltage and find the turn ratio.

APPARATUS:

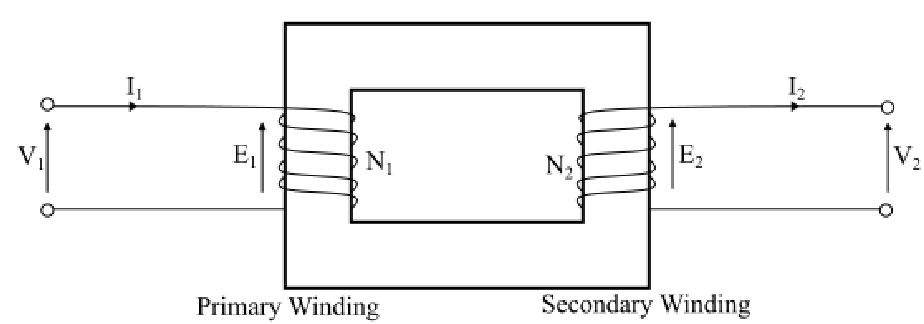
S.NO.	ITEMS	RATING	QUANTITY
1.	1 Phase Transformer	1 KVA, 1 Phase, 230 V, 50 Hz	01
2.	Volt meter	Analog Voltmeter (0-230-600V)	02
3.	Ammeter	Analog, MI, (0-5-10A)	02

THEORY

Transformers are magnetic circuit elements used in AC circuits for a variety of applications. Transformers are often used to convert voltages from high to low values, or from low to high values. They are often used to isolate one AC line from another, for safety or for equipment isolation. And they are often used to connect two components with mis-matched impedances, such as a high-impedance stereo amplifier and a low-impedance speaker. In its simplest form, a transformer consists of two closely coupled wire coils that share a common magnetic field. If a voltage is applied to one of the coils (called the primary), then a voltage will be induced on the second coil (called the secondary). The actual voltage induced in the secondary coil is proportional to the ratio of the number of turns in the two coils. The *turn ratio* of a single phase transformer is defined as the ratio of number of turns in the primary winding to the number of turns in the secondary winding, i.e. That ratio is called the turns ratio, for which we will use the symbol a. If the number of turns in the primary is N1 and the number of turns in the secondary is N2, the turns ratio a is

$$\mathbf{a = N_2/N_1}$$

$$\mathbf{a = N_2/N_1 = V_2/V_1}$$

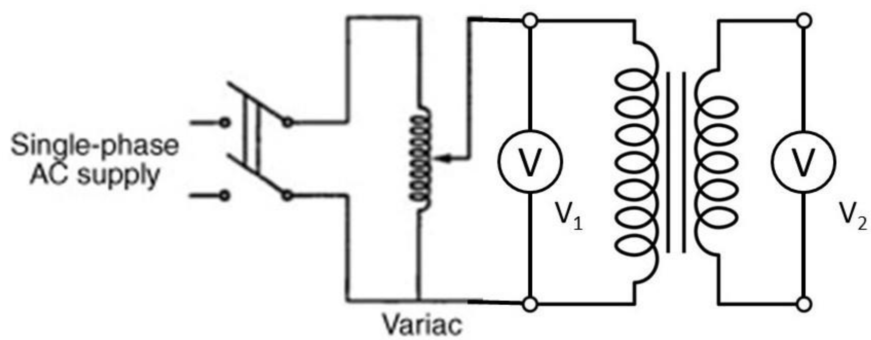


The two coils (also called *windings*) may be wound around air (an *air core*) or a dielectric, but more often they are wound around a ferromagnetic material, such as steel (an *iron core*), in order to concentrate the magnetic field and improve the coupling between the coils.

PROCEDURE

Turn Ratio Test:

- Connect the circuit as shown in the diagram.
- Switch on the a.c. supply.
- Record voltage V1 across primary and V2 across various tappings of secondary.
- If V1>V2 then transformer is step down.
- If V2> V1 then transformer is step up.
- Switch off a.c. supply



RESULT

Thus we have studied about transformer connection and performed turn ratio test.

VIVA QUESTIONS

Q1 what is transformer?

Ans. Transformer is a static device which is used to change the level of voltage or current without changing the frequency and power.

Q2 what do you mean by turns ratio of transformer?

Ans. Turns ratio of a transformer is the ratio of primary turns to the secondary turns.

Q3 what is transformation ratio of transformer?

Ans. Transformation ratio is the ratio of secondary side turns to primary side turns.

Q.4 what are the application of an autotransformer?

Ans. Generally we use autotransformer to get variable voltage output. Many electrical applications required variable input.